MA329 Statistical linear models 25-26

Assignment 1 (Due date: Sept 30, 11pm. For late submission, each day costs 10 percent)

1. (30 marks)
   (a) Define a simple linear regression model and derive MLE (maximum likelihood estimation) for all the unknown parameters
   (b) Comments on the difference between MLE and LSE (least square estimation)
   (c) Under the model given in (a), prove that $\bar{y}$ and $\hat{\beta}_1$ are independent (using the notation given in the lectures).

2. (50 marks, don't use R outcome directly) The following table gives the data on daytime eruption of OLD Faithful Geyser in Yellowstone National Park on August 1, 1978. The variables are x =duration of an eruption and y =interval to the next eruption (both in minutes).

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$x$  4.4 3.9 4.0 4.0 3.5 4.1
$y$   78  74  68  76  73  84
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   (a) Construct and comment a scatterplot of the data.
   (b) Find the least squares line from the data and plot it on your scatterplot.
   (c) What is your linear regression model? State the necessary assumptions.
   (d) Test the hypothesis that the duration of an eruption has no effect of the interval to the next eruption when a linear model is used (use $\alpha = 0.05$). State the null and alternative hypotheses. Draw the appropriate test conclusions.
   (e) Find a 95% confidence interval for β_1 (the slope of the linear regression model). Interpret your results.
   (f) Find the coefficient of determination for the linear regression model. Interpret your result.
   (g) Find a prediction of the time to the next eruption when the Geyser eruption lasts for 4 minutes and its 95% interval.